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MILITARY STANDARD

MACROGRAPH STANDARDS FOR STEEL BARS, BILLETS, AND BLOOMS FOR AMMUNITION COMPONENTS

FSC 9640

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DEPARTMENT OF DEFENSE

Washington, D. C.

Macrograph Standards for Bars, Billets, and Blooms for Ammunition Components

1. This Military Standard is approved for use by all Department of the Army Activities.

2. Recommended corrections, additions, or deletions should be addressed to Commander, Frankford Arsenal, ATTN: SARFA-J4000, Philadelphia, Penna., 19137.

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MACROGRAPHS

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1. SCOPE

1.1 This standard provides macrographs of physical defects and chemical nonhomogeneity of steel bars, billets, and blooms up to 100 square inches maximum cross sectional area. It is intended for use with carbon and alloy steels. This standard is not applicable to resulphurized steel.

2. REFERENCED DOCUMENTS

STANDARDS

Federal

Fed. Test Method Std. No. 151-Metals;
Test Methods

3. DEFINITIONS

Not applicable.

4. GENERAL STATEMENTS OR REQUIREMENTS

4.1 Description of macrographs

4.1.1 Four classifications of defects as follows, are indicated by the macrographs in this standard.

(a) Series A - Center defects
(b) Series B - Subsurface defects
(c) Series C - Ring defects
(d) Series D - Miscellaneous defects

4.1.2 In each series designated by the letters A, B, and C the macrographs demonstrate the same classification of defect but with a different degree of severity; the seriousness of the defect increasing with the increasing numerical designation. Series D covers miscellaneous defects. Each macrograph in this series is an independent illustration of a particular type of defect.

4.1.3 All of the macrographs in this standard are full size.

4.2 Method of specifying. The macrographs do not indicate limits of acceptability but provide a means of systematically specifying limits in detail specifications and contracts or purchase orders for steel bars, billets, and blooms. In prescribing limits, the intended use of the material and the probable effect of each type of defect should be kept in mind; and the limits should represent the minimum quality suitable for the purpose. The maximum allowable defect of each classification, except series D, should be specified by reference to the series letter and the number of the photo-macrograph in-

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dicating the most severe defect which shall be acceptable. Each unacceptable miscellaneous defect shown in series D should be specified by reference to the series letter and the number of the macrograph indicating the defect. To avoid confusion, it is important that the method of specifying the macroetch requirements be patterned on the following form, which is intended to be merely an example rather than an indication of which defects should be considered rejectable in any particular case.

Example: The quality and cleanliness of the steel as indicated by the results of the macroetch test, shall be equal to or better than macrographs A5-B4-C8, with defects D3, D4, D6, D7, and D8 unacceptable.

5. DETAILED REQUIREMENTS

5.1 Test specimens. The number and location of test specimens shall be as specified in the detail specification or contract or purchase order.

5.2 Test procedures. Test procedures and preparation of test specimens shall be in accordance with Method 321 of Federal Test Method Standard No. 151.

5.3 Use of macrographs. The steel shall meet the requirements for soundness and freedom from defects specified in the applicable specification, contract, or purchase order by reference to the macrographs contained herein. The presence in any macroetch specimen of any defect or defects of greater severity than indicated by the specified series A, B, and C macrographs shall be grounds for rejection. The presence of any defect of the type indicated by the specified series D macrographs shall also be grounds for rejection.

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DOCUMENT IDENTIFIER AND TITLE

MIL-STD-1459A(MU) Macrograph Standards for Steel Bars, Billets, and Blooms ~~for Ammunition Components~~

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
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2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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